

Astronauts Stay Busy on International Space Station Awaiting Flight Home

Two American astronauts awaiting their return flight from the International Space Station (ISS) have stayed busy carrying out a series of scientific experiments.

The U.S. space agency NASA has said the pair – Butch Wilmore and Suni Williams – will remain at the ISS until late March or early April. When they arrived at the orbiting laboratory in June 2024, the astronauts expected to stay in space for just one week.

Their trip was a test flight of Boeing's Starliner spacecraft. While Starliner safely transported the two to the ISS, the spacecraft had several problems that persuaded NASA to send it back to Earth for further testing without the astronauts onboard.

NASA's current plan calls for Wilmore and Williams to fly back on a SpaceX spacecraft in late March at the earliest. NASA has been using SpaceX's Crew Dragon to regularly carry astronauts and supplies to the ISS since 2020.

NASA astronaut Suni Williams displays portable carbon dioxide monitors aboard the International Space Station. (NASA)

The astronauts have been using their extended mission to carry out experiments. In addition, Williams stepped out of the space station earlier this month to complete a spacewalk.

Another NASA astronaut, Nick Hague, joined Williams on the spacewalk on January 16, the agency said in a statement.

The two repaired a piece of equipment that helps guide the movements of the ISS. The spacewalk also included work on an X-ray telescope, as well as maintenance activities for other ISS systems. NASA noted it was the fourth spacewalk for Hague and the eighth for Williams.

The space agency said Williams was preparing for another spacewalk planned for January 30. Wilmore is expected to join Williams on that spacewalk, which NASA estimates will last about six-and-a-half hours. The two will be aiming to remove a radio antenna structure. They will also be "searching for microbes" around the ISS for use in experiments.

Here is a look at other experiments Wilmore and Williams have completed during their stay:

Astronauts Butch Wilmore and Suni Williams prepare orbital plumbing equipment inside the International Space Station. (NASA)

New methods to grow food

NASA said Wilmore and Williams have tested different methods for growing crops in space. Such methods could support humans who have longer stays in space in the future.

One experiment is called the Plant Water Management investigation. NASA said this experiment “uses the physical properties of fluids...to overcome the lack of gravity and provide hydration to plants.” The experiment aims to help NASA develop hydroponic systems to support future space missions.

Another experiment examines a new way to grow vegetables, plants and flowers inside the ISS to provide nutrition and beauty. As part of this experiment, Wilmore installed a new light meter inside the ISS. The device is designed to measure available light and to help crew members improve growing operations.

NASA astronaut Suni Williams installs the Packed Bed Reactor Experiment inside the International Space Station's Microgravity Science Glovebox. (NASA)

Robot assistants

NASA also said that throughout the current mission, Williams had deployed and observed three free-flying robots called Astrobees. This robotic experiment seeks to examine how such robots could assist astronauts with daily duties and maintenance work.

The agency said the square-shaped robots are designed to help scientists and engineers develop and test new robotic technologies for use in microgravity environments. The assistants are also designed to help astronauts with daily tasks “and give ground controllers additional eyes and ears on the space station.”

Williams tested the robots’ ability to map the inside of the space station and Astrobees’ flight abilities.

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Image captures from space

NASA astronauts visiting the ISS have a long history of capturing images of Earth from space. And this is another activity Wilmore and Williams have carried out. The ISS gives astronauts a “window to the world” as they orbit 400 kilometers above their home planet.

During their mission, NASA said the two astronauts had “captured hundreds of photographs of Earth, ranging from auroras, land, sea, orbital sunrises and sunsets, and more.”

Radio connections with Earth

Another space station tradition is for U.S. astronauts to connect with students and other interested individuals during their stay in space. They have virtual visits with people back on

Earth who can ask the astronauts how it feels to live in space and can learn about NASA's experiments aboard the ISS.

NASA astronaut Butch Wilmore performs spacesuit maintenance inside the International Space Station's Quest airlock compartment. (NASA)

In early August, Williams used ham radio technology to connect with students from Banda Aceh, Indonesia. NASA explained that students get the chance to communicate with the astronauts when the ISS passes overhead. The students can ask about research on the ISS as well as the radio wave technology that makes the communication possible.

I'm Bryan Lynn.

Bryan Lynn wrote this story for VOA Learning English, based on reports from NASA and The Associated Press.

Quiz - Astronauts Stay Busy on the International Space Station Awaiting Flight Home

Vocabulary

antenna - n. อุปกรณ์ที่ใช้ส่งหรือรับสัญญาณการสื่อสาร

hydration - n. กระบวนการที่ทำให้ร่างกายได้รับน้ำหรือของเหลวอื่นๆ

hydroponic - adj. เกี่ยวข้องกับการหรือปลูกโดยใช้วิธีการปลูกพืชในน้ำ ทราย หรือกรวด

install - v. ติดตั้ง (อุปกรณ์)

meter - n. เครื่องวัด

maintenance - n. การบำรุงรักษา

aurora - n. แสงออโรรา

virtual - adj. เสมือนจริง

ham radio - n. วิทยุสมัครเล่น

Reading Comprehension Quiz

1. What is the primary reason the two American astronauts are currently at the International Space Station (ISS)?

- A) They are on their scheduled return flight.
- B) They are conducting scientific experiments.
- C) They just arrived for a new mission.
- D) They are assisting with spacecraft repairs.

2. What was the initial expected duration of the astronauts' stay at the ISS when they first arrived?

- A) Until late March or early April.
- B) Several months, starting June 2024.
- C) Just one week.
- D) Indefinitely due to unforeseen issues.

3. Why was Boeing's Starliner spacecraft sent back to Earth without the astronauts onboard?

- A) It was its scheduled return trip after the test flight.
- B) NASA decided to use a different spacecraft for the return.
- C) The Starliner experienced several problems requiring further testing.
- D) The astronauts opted to extend their stay on the ISS.

4. According to NASA's current plan, which spacecraft is expected to transport Wilmore and Williams back to Earth?

- A) Boeing's Starliner spacecraft.
- B) A new, unnamed NASA spacecraft.
- C) SpaceX's Crew Dragon spacecraft.
- D) The International Space Station itself.

5. What additional activity did Suni Williams complete earlier this month, besides carrying out experiments?

- A) She returned to Earth for medical check-ups.
- B) She welcomed new astronauts to the ISS.
- C) She performed a spacewalk outside the station.
- D) She trained new robots for daily duties.

6. Who accompanied Suni Williams on her spacewalk on January 16?

- A) Butch Wilmore.
- B) An unnamed space agency technician.
- C) Nick Hague.
- D) A robot assistant.

7. What was one of the primary tasks performed during the spacewalk mentioned in Paragraph 9?

- A) Installing a new radio antenna structure.

- B) Searching for microbes around the ISS.
- C) Repairing equipment that guides ISS movements.
- D) Testing a new hydroponic system.

8. What is one of the main objectives of the spacewalk planned for January 30?

- A) To install new scientific instruments.
- B) To remove a radio antenna structure.
- C) To repair damaged solar panels.
- D) To conduct a survey of external damage.

9. Why are the astronauts testing new methods for growing crops in space?

- A) To provide fresh food for their current mission.
- B) To support future long-duration human space missions.
- C) To determine the effects of microgravity on plant growth.
- D) To cultivate plants for aesthetic purposes only.

10. The Plant Water Management investigation aims to develop which type of system for future space missions?

- A) Advanced air filtration systems.
- B) Recycled waste management systems.
- C) Hydroponic systems.
- D) Robotic farming equipment.

11. What is the purpose of the new light meter installed by Wilmore as part of a growing experiment?

- A) To detect harmful radiation levels.
- B) To measure available light for improving plant growth.
- C) To regulate the temperature inside the growing chambers.
- D) To monitor water levels in hydroponic systems.

12. What is the primary focus of the Astrobees robotic experiment?

- A) To develop new navigation systems for spacecraft.
- B) To test their ability to perform external repairs on the ISS.
- C) To examine how robots can assist astronauts with daily duties and maintenance.
- D) To map unknown areas of the solar system.

13. Besides assisting astronauts with daily tasks, what other benefit do the Astrobees robots provide according to Paragraph 20?

- A) They act as personal companions for astronauts.
- B) They offer ground controllers 'additional eyes and ears' on the space station.
- C) They generate electricity for the ISS.
- D) They autonomously conduct complex scientific experiments.

14. What historical activity, also performed by Wilmore and Williams, is mentioned as a tradition for NASA astronauts visiting the ISS?

- A) Conducting spacewalks for maintenance.
- B) Capturing images of Earth from space.
- C) Communicating with international space agencies.
- D) Testing new propulsion systems.

15. Which of the following natural phenomena were among the 'hundreds of photographs of Earth' captured by the astronauts?

- A) Volcanic eruptions.
- B) Lightning storms.
- C) Auroras.
- D) Meteor showers.